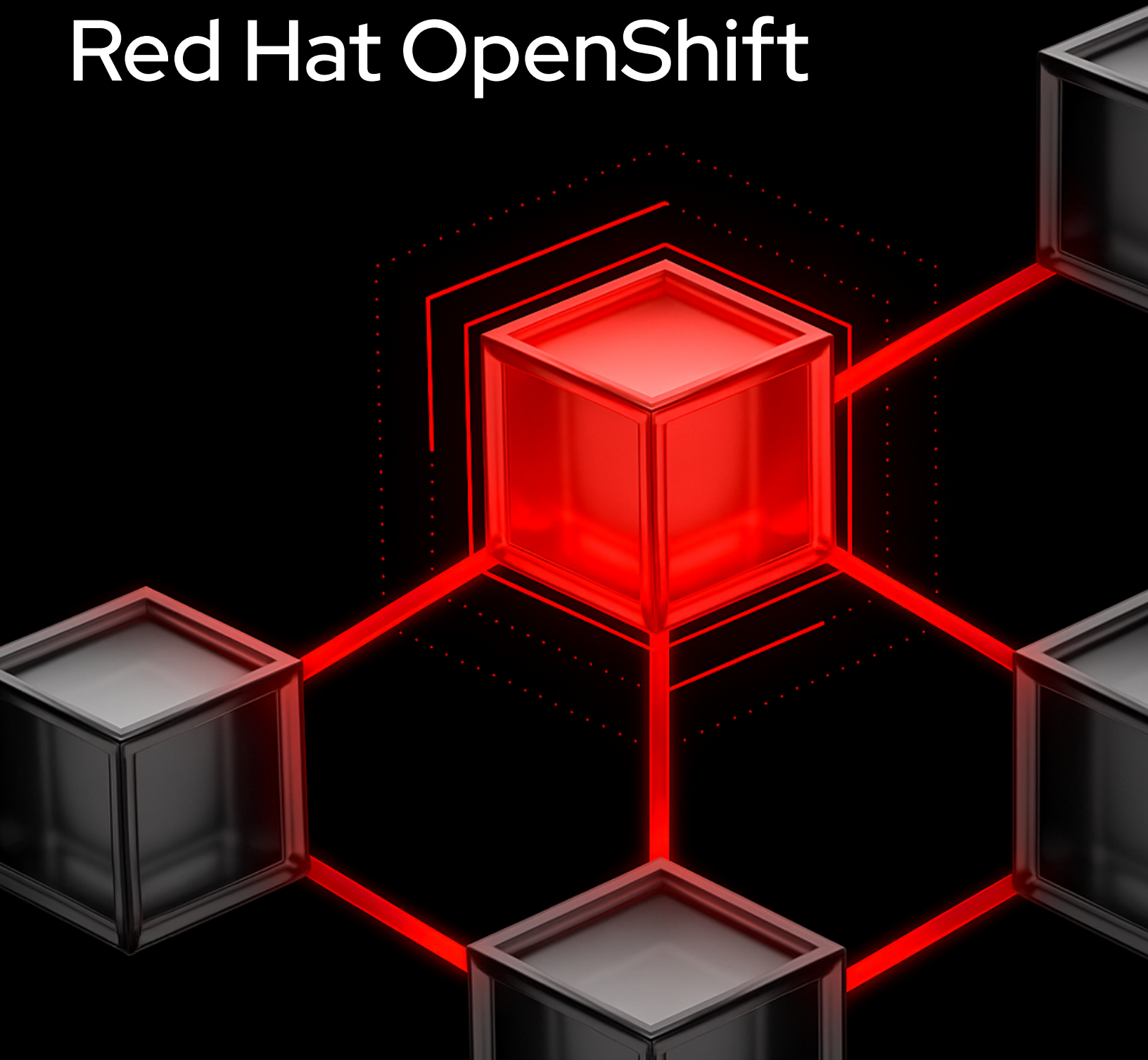
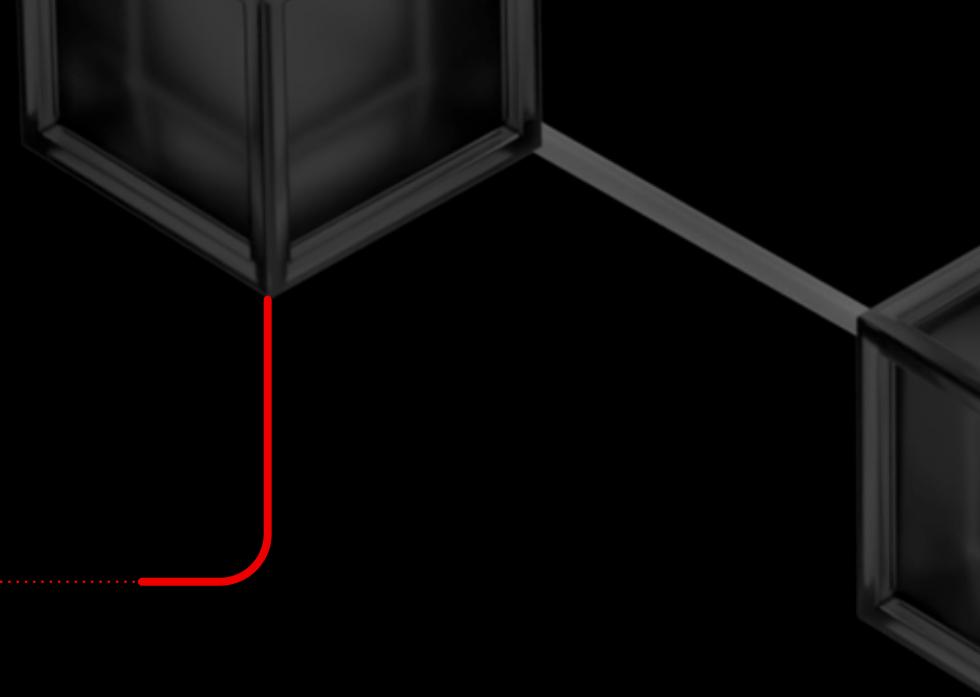


15 reasons to modernize your virtual machines to Microsoft Azure Red Hat OpenShift



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Virtualization infrastructure continues to evolve



For decades, organizations have relied on virtualization technologies to deliver innovative applications and services that streamline operations, optimize costs, and promote new business opportunities.

The introduction of virtual machines (VMs) into datacenters helped IT teams efficiently use resources, improve flexibility, and optimize infrastructure performance. As public clouds became more accessible, virtualization platforms evolved to make use of the scalability, agility, and potential cost benefits offered by new cloud computing models.

Now, many organizations are reassessing their virtualization platforms and strategies to make sure they have the right combination of consistency, efficiency, and support for their future operations and cloud-based applications. These organizations want to rapidly migrate to unified, cost-effective platforms that can handle their existing VMs while preparing them for IT modernization. Even so, many enterprises will continue to use virtualization technologies well into the future to maximize their return on investment (ROI) and to schedule operational changes when it's least disruptive to business.

Modern, cloud-native application platforms that support both traditional VMs and lightweight technologies such as containers across hybrid and managed cloud environments are of particular interest for organizations on a journey to modernization. Teams can readily migrate their current VMs to the new platform and run them alongside container-based applications to take advantage of new approaches to building and deploying applications.

At the same time, container orchestration platforms often include tools and automation that help IT teams streamline lifecycle management of applications across massive hybrid cloud environments. IT teams can deploy and administer all workloads—virtualized or containerized—using common processes and tools. They can also take advantage of new innovation in monitoring, development and deployment pipelines, GitOps, service meshes, and serverless technologies. As a result, organizations can continue to run the virtualized workloads their business relies on while modernizing their applications and approaches at their own pace.

To be successful, IT organizations need a cloud-ready application foundation that meets their current virtualization needs while preparing them for future modernization and change. [Red Hat® OpenShift®](#) provides a modern, enterprise-ready platform for running VMs and container-based applications. Powered by containers, Kubernetes, and DevSecOps capabilities, Red Hat OpenShift is a security-focused foundation for rapidly building, deploying, running, and managing both existing and new applications at scale across hybrid, multicloud, and edge environments.

Jointly engineered, operated, and supported by Red Hat and Microsoft, [Microsoft Azure Red Hat OpenShift](#) gives you a fully managed, ready-to-use application platform. This cloud service combines Red Hat OpenShift, Azure infrastructure, and an expert site reliability engineering (SRE) team to deliver a reliable foundation for innovation.

Azure Red Hat OpenShift offers more than just access to a managed application platform. It provides a complete, full-stack environment with all necessary services and technologies, simple self-service capabilities, and expert 24x7 support with a 99.95% availability service level agreement (SLA). SRE teams use proven best practices and extensive experience to build, operate, automate, and maintain your Azure Red Hat OpenShift deployment.

As a first-party cloud service, Azure Red Hat OpenShift can be accessed on-demand from the Azure console and integrates with other Azure cloud-native services such as Azure OpenAI, Azure Arc, and more. You can choose hourly billing or reserved instances, receive a single invoice for Azure Red Hat OpenShift and Azure deployments, and use Azure committed spend and discount programs for Azure Red Hat OpenShift.



Expert platform and cloud management

With Azure Red Hat OpenShift, Red Hat and Azure experts manage the entire software stack, including:



Cluster creation.



Cluster management.



Monitoring and logging.



Network configuration.



Software and security updates.



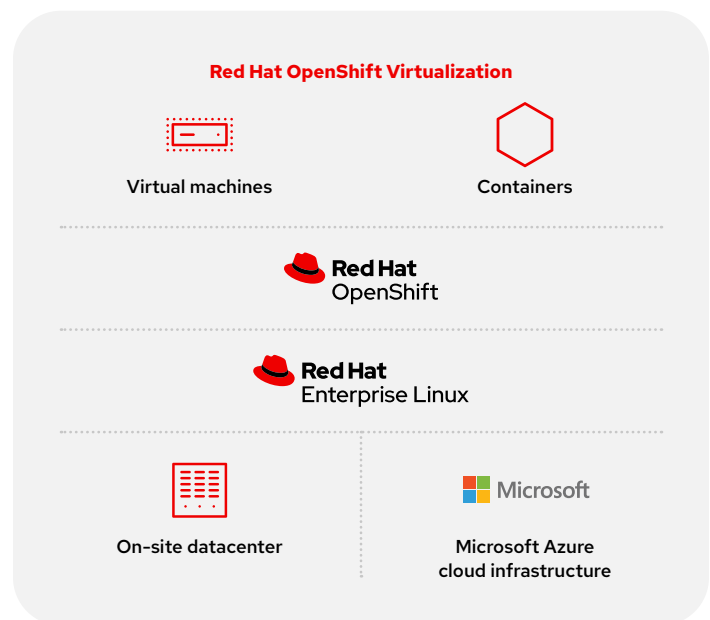
Platform support.

Migrate your virtual machines to a cloud-ready virtualization platform

Included as a feature of Azure Red Hat OpenShift and based on the Kernel-based Virtual Machine (KVM) and KubeVirt open source projects, [Red Hat OpenShift Virtualization](#) lets you migrate your Linux® and Microsoft Windows VMs to a unified application platform, at your own pace.

With the assistance of [Red Hat Ansible® Automation Platform](#) on Microsoft Azure (available as a managed offering), you can move VMs from other platforms and run them on Azure Red Hat OpenShift. This lets you get the most from your existing virtualization and cloud spending investments while taking advantage of cloud-native architectures, streamlined operations and management, and new development approaches.

You can run OpenShift Virtualization as a self-managed operator on Azure Red Hat OpenShift. Red Hat and Azure SRE teams manage the underlying Red Hat OpenShift platform and Azure infrastructure as usual, while your internal teams manage OpenShift Virtualization and the VMs running on it. Red Hat experts remain available to answer questions and support your teams as they use OpenShift Virtualization.



Migrating your VMs to Azure Red Hat OpenShift delivers value at every point in the application modernization journey.

Gain benefits across your organization

Here are 15 reasons to migrate your traditional VMs to a unified foundation based on Azure Red Hat OpenShift.

1

Maintain stability and support on a modern virtualization platform

Migrating workloads from traditional to modern virtualization platforms can be complicated. To make sure the migration process is smooth and efficient, you need to carefully consider VM compatibility, possible configuration changes, and potential performance optimizations. Additionally, differences in virtualization platform deployment and management processes and tools need to be accounted for in your migration plans, especially when moving between cloud providers. Preemptively validating VM compatibility, using warm migration capabilities, and transitioning multiple VMs at once can help you move workloads between virtualization platforms more rapidly and with less effort.

The [migration toolkit for virtualization](#) streamlines and accelerates the process of moving existing VMs to OpenShift Virtualization, saving time and minimizing potential errors. Plus, by combining this toolkit with Ansible Automation Platform, you can [automate the migration](#) of groups of VMs and related infrastructure—such as networking and storage—at scale and speed.



Operate consistently and cost-effectively across hybrid and multicloud environments

One of the key advantages of adopting hybrid and multicloud environments is flexibility. During application deployment, these environments let you choose between various datacenter and cloud resources to balance scalability, performance, and cost according to your business objectives. Application platforms that run and migrate VM workloads consistently across hybrid and multicloud cloud environments help you streamline application deployment, optimize resource use, and maintain operational consistency.

OpenShift Virtualization supports self-managed physical servers in datacenters, edge environments, and public cloud environments, so you can deploy and manage your VMs on optimal infrastructure. Because Azure Red Hat OpenShift runs consistently in datacenter and public, private, or hybrid cloud environments, you can choose the right location for each application and VM, and move them as needs change. This consistency also streamlines the migration of on-site workloads and VMs to Azure.

Managing separate platforms for VMs and containers can lead to increased complexity, resource fragmentation, and operational overhead. Unified platforms that streamline infrastructure deployment, management, and monitoring across both VMs and containers can help you optimize resource use, eliminate duplicate efforts, and quickly adapt to diverse workloads. [Red Hat Advanced Cluster Management for Kubernetes](#) helps Red Hat OpenShift Virtualization scale by providing a single console to manage VMs and containers across clusters, with built-in governance, policy enforcement, and automation.

OpenShift Virtualization streamlines operations with a single platform for VMs, containers, and bare-metal and serverless workloads. Standardize infrastructure deployment and maintain all workloads using a common, consistent set of established, enterprise tools. You can also streamline Day 2 operations from provisioning to compliance to deprovisioning with Ansible Automation Platform.

Plus, you can continue to use your existing services and applications with Azure Red Hat OpenShift via certified partner integrations. Deploy and manage your preferred applications more easily and confidently with [certified operators](#) and Helm charts that encompass detailed vendor expertise. For example, Azure offers a variety of [operators for managing Azure services](#) from within Red Hat OpenShift.



3

Optimize cloud cost

As a first-party service, Azure Red Hat OpenShift can be provisioned directly from the Azure console and integrates with other Azure cloud-native services such as Azure OpenAI and Azure Arc. This makes it easier to unify development and operations under single environment without adding extra management overhead or third-party complexity.

Azure Red Hat OpenShift gives organizations multiple ways to optimize spending while keeping operations streamlined and predictable. With flexible payment models, you can align costs with your business goals: hourly billing scales up or down with demand, while reserved instances provide stability and predictable budgeting over time. Azure Red Hat OpenShift also streamlines procurement with a single invoice for both Red Hat OpenShift and Azure deployments, and you can apply existing Azure committed spend or discount programs to maximize negotiated savings.

Using hardware overcommit, organizations can run more VMs using fewer instances, and greatly reduce resource allocation.

The combination of flexible payment options, straightforward billing, and infrastructure-level efficiency provides a comprehensive path to balancing agility with cost control.



4

Modernize your applications over time, on your schedule

While migrating monolithic or n-tier applications on VMs to containerized, microservices-based workloads can deliver major benefits, enhancing scalability, productivity, and agility, it can also require significant investments in time and resources. That's why platforms that support mixed environments, including VMs, containers, bare metal, and even serverless workloads, are critical. They let you strategically transform applications at your own pace and according to your unique business needs.

With OpenShift Virtualization on Azure Red Hat OpenShift, you gain a single, unified platform to modernize without disruption. You can run VM- and container-based workloads side by side, integrate cloud-native applications and AI workloads, and maintain operational consistency across on-premise, cloud, and edge environments. This reduces the burden of ongoing platform and resource management, freeing teams to focus on strategic innovation, upskill in emerging technologies such as gen AI, and accelerate AI-enabled application development through the combined Azure and Red Hat OpenShift ecosystems.

Because Azure Red Hat OpenShift development and operations are unified in a single environment, it can be provisioned directly from the Azure console and integrates with services such as Azure OpenAI and Azure Arc. This eliminates extra management overhead or third-party complexity while providing consistency, scalability, and a faster time to value for modern workloads.



Transform your applications

Red Hat OpenShift runs consistently across on-site datacenters, cloud environments, and managed cloud services, so you can choose the optimal environment for your applications and VMs.

[Read the e-book](#)



With Azure Red Hat OpenShift, you can build an ideal software foundation for all application transformation use cases using a complete, integrated portfolio of products and cloud services that work together reliably. Directly access and integrate other Azure cloud-native services for compute, database, analytics, machine learning (ML), networking, mobile, and more, including:

- [FinOps on Azure](#). Cloud cost management and optimization solutions.
- [Azure databases](#). A set of fully managed, purpose-built databases.
- [Azure machine learning](#). A comprehensive set of AI and ML services, infrastructure, and resources.
- [Azure governance](#). A set of business management and governance control services.
- [Azure Content Delivery Network](#). A broad set of services for running workloads and delivering content over a reliable, security-focused network.
- [Azure security tools and services](#). A broad set of identity and access management, threat detection, compliance, and network, application, and data protection services.
- [Azure storage services](#). A portfolio of solutions for storing, accessing, protecting, and analyzing your data.

5



Provide self-service options for deploying VMs

Manually deploying VMs is an inefficient, error-prone process that can result in inconsistent configurations, long deployment times, and an increased risk of security vulnerabilities. Self-service capabilities let users rapidly and reliably deploy preapproved, security-compliant VM configurations, when they need them and without opening an IT service ticket.



With Azure Red Hat OpenShift, users can quickly and independently provision the resources they need without manual intervention from IT operations teams. Following standard Red Hat OpenShift role-based access controls (RBAC), users can create VMs in their projects—and then grant access to other project members—to help their entire team quickly get the resources they need. [VM instance types](#) streamline self-service provisioning via predefined operating system (OS) images, workload types, and hardware requirements. You can also use [templates](#) to deploy VMs that require advanced configuration, including virtual appliances.

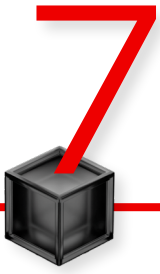
6



Integrate VMs into your development and deployment pipelines

Using VMs in development and deployment pipelines can increase the scalability, consistency, and speed of your application delivery processes. Integrating VMs in development pipelines lets you deploy standardized, isolated, and reproducible environments for coding, testing, and debugging to increase consistency across development teams. Using VMs in [continuous integration/continuous deployment \(CI/CD\)](#) pipelines provides clean, isolated environments for each stage—from build and test to release and deployment—for controlled, reliable application delivery.

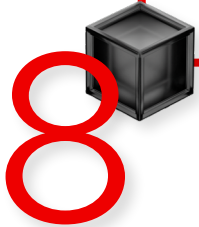
With OpenShift Virtualization, you can create, manage, and run commands in VMs within development and CI/CD pipelines based on [Red Hat OpenShift Pipelines](#) to streamline infrastructure and application delivery. Azure Red Hat OpenShift also gives you access to a large library of integrated, certified tools, development environments, and self-service capabilities that let developers code at speed.



Take advantage of production-ready virtualization hypervisor technologies

Hypervisor performance, stability, and security posture are critical for efficient, dependable virtualization infrastructures. Adopting extensively tested and validated hypervisors that are supported by trusted vendors can help you better manage virtualized workloads at scale and increase reliability across diverse environments.

As the underlying hypervisor for OpenShift Virtualization and Red Hat Enterprise Linux, [KVM](#) is a security-focused, high-performance, open source hypervisor. First released in 2007, KVM provides a stable, efficient virtualization foundation for organizations worldwide. Today, Linux virtualization powers critical IT infrastructure for a large number of global financial services firms, airlines, manufacturers, public sector organizations, and telecommunications companies and is a popular choice for public cloud deployments.

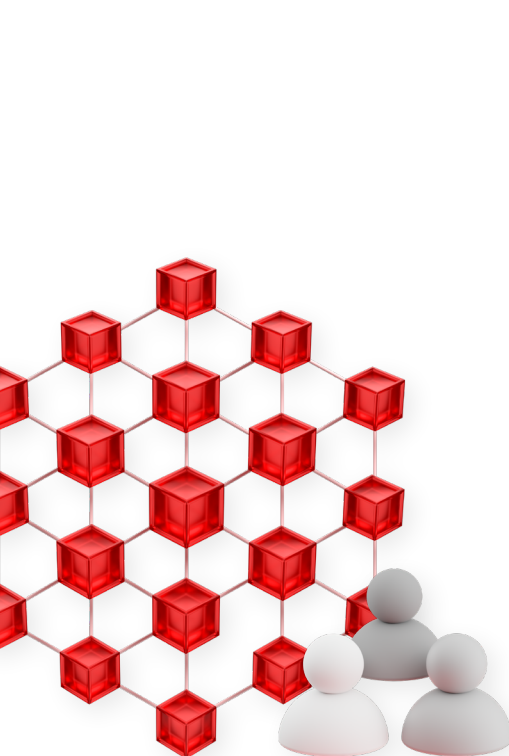


Boost VM performance

Fast recovery times are critical when IT services experience interruptions that cause downtime. When this occurs, applications running in VMs that use those services are also unavailable. An application platform that can recover and reboot VMs quickly and efficiently is essential for keeping your business up and running at all times.

OpenShift Virtualization exhibits near-linear boot times for large numbers of VMs, so your critical applications can always be available.





9

Accommodate multiple guest OSes

In virtualized environments, support for guest OSes increases the diversity of workloads, applications, and services you can run on shared physical infrastructure. Compatibility with a wide range of OSes, advanced security features that isolate guests and hosts, and support from experts with extensive experience simplifies virtualization across varied IT environments.

Red Hat tests, certifies, and supports [guest OSes](#) for use with OpenShift Virtualization—including certification for Microsoft Windows guest support through Microsoft’s Server Virtualization Validation Program (SVVP)—to help you create an IT environment that meets your business needs. These OSes, including Windows Server, Red Hat Enterprise Linux, or other versions of Linux, can be managed through Ansible Automation Platform. You can also continue to use common in-guest tools such as PowerShell and Ansible Automation Platform with VMs running on OpenShift Virtualization.

10

Decrease risk with advanced security features and best practices

Security vulnerabilities in virtualized environments with shared hardware infrastructure increase the risk of unauthorized access, data breaches, and potential service disruptions. Strong isolation technologies, consistent security policies, and adherence to least privilege principles enhance the overall security posture of VM workloads.

OpenShift Virtualization follows the restricted Kubernetes pod security standards profile and runs VM workloads without root privileges, helping you comply with current, industry-standard security practices and protect your organization. Red Hat OpenShift also includes core security features—such as access controls, network security, and an enterprise registry with a built-in scanner—to safeguard your platform from the start.

Azure Red Hat OpenShift is certified to and managed in compliance with [key security standards](#), including Payment Card Industry Data Security Standard (PCI DSS), System and Organization Controls (SOC) 2, and International Organization for Standardization (ISO) 27001.

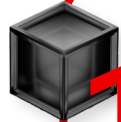
In fact, Azure infrastructure includes many security features and more than 50 compliance certifications and accreditations to help you protect your organization and assets. You can integrate other security and access management services—such as [Red Hat Advanced Cluster Security for Kubernetes](#), [Azure management and governance tools](#), and [Azure security tools and services](#)—to further safeguard your business and IT assets.

11

Simplify live migration of VMs

Live migration—moving a running VM to another host without interrupting the workload—is crucial for maintaining continuous operations as infrastructure demands change. A virtualization platform that lets you configure, initiate, and monitor live migrations across your environment helps balance workloads and avoid downtime during maintenance activities.

OpenShift Virtualization supports complete [live migration](#) workflows with a unified management console, configurable policies, VM metrics, and traffic encryption to help you keep your applications running reliably. You can migrate live VMs between compute instances, storage instances, and even Azure availability zones.



12

Back up and restore VMs

When unexpected events or system disruptions occur, backup and restore capabilities can help you recover VMs in less time and make sure operations continue. Creating frequent backups—representations of the state and data of a VM at a specific point in time—provides the information needed to restore existing VMs.

OpenShift Virtualization lets you automatically [back up VMs](#) on demand or on fixed schedules, manage saved images, and restore workloads quickly to minimize the effects of disruptions on your business. Red Hat's certified partner ecosystem also includes many third-party products for [data storage, backup, and restoration](#). Using the Red Hat OpenShift Operator Framework, you can interact with many of these products directly from Red Hat OpenShift. For example, you can install and configure Ansible Automation Platform via an operator, and use the platform to automate VM lifecycle management, including backup and restore operations.

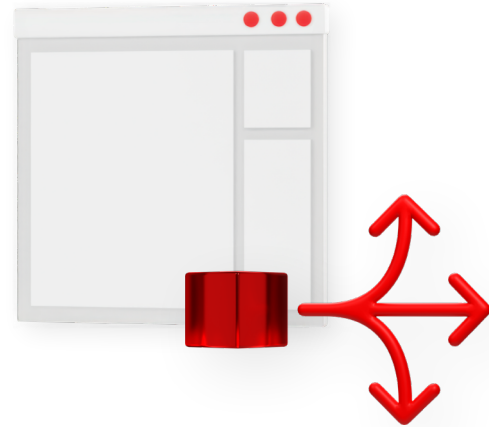


13



Scale infrastructure as workloads change

The dynamic workloads, diverse technologies, and rapid pace of development and deployment associated with modern applications places high demands on IT infrastructure. To deliver optimal performance and resource use, virtualization platforms must dynamically and efficiently scale up and down as workloads change. Ansible Automation Platform can assist with this by helping with the scaling of cluster resources and orchestrating processes such as load balancing and configuration.



[Machine management](#) features in Red Hat OpenShift—including autoscaling based on workload policies and machine health checks—help you administer infrastructure with greater flexibility and efficiency to meet modern application demands. And with Azure Red Hat OpenShift, Red Hat’s SRE team monitors your control plane usage 24x7 and will notify you if a cluster needs to be scaled to avoid an outage.

14



Support a collaborative, open source model

Open source development models promote collaboration, innovation, and community-powered development to rapidly deliver new, advanced virtualization technologies. With access to stable community innovation, open standards for broad compatibility, and open application programming interfaces (APIs) for flexible integration, open source technologies can help you build efficient virtualized environments across datacenter and cloud infrastructures.

OpenShift Virtualization uses container-native virtualization technology to deliver ongoing innovation. This technology is developed and maintained in [KubeVirt](#), a Cloud Native Computing Foundation (CNCF) project. As the foundation of Red Hat OpenShift Virtualization, KubeVirt provides a unified development platform where developers can build, modify, and deploy applications residing in both containers and VMs in a common, shared environment.



Work with our cloud and virtualization experts

Successfully planning, deploying, and maintaining a virtualized environment requires specialized skills and knowledge. Expert support and guidance, backed by extensive virtualization experience and in-depth platform knowledge, can help you optimally configure your environment, proactively resolve potential issues, and maximize performance, security, and reliability.

Azure Red Hat OpenShift helps you optimize support costs and operations with included 24x7 expert support. Red Hat's SRE team automates the deployment and management of your clusters so your teams can focus on application development and strategic initiatives. Reduce costly downtime and maintain reliability and security with managed threat monitoring and

remediation. Resolve issues in less time via a single path to support through Red Hat or Azure. Access Kubernetes experience without retraining or moving existing staff or hiring new members.

Although your teams are responsible for managing OpenShift Virtualization on top of Azure Red Hat OpenShift, Red Hat SRE and support staff are still available to answer questions and provide expertise about migrating, running, and managing VMs on the platform.

You can also accelerate deployment with migration planning assistance through [Red Hat Consulting](#) engagements.

A consistent foundation for innovation

By giving you a single scalable and flexible platform for VMs and containers, OpenShift Virtualization reduces operational overhead and opens a straightforward path to modernization. This integration offers a unified approach for managing VMs and containers in an efficient, security-focused manner. And with Azure Red Hat OpenShift, you can take advantage of a unified foundation, integrated products and services, a large partner ecosystem, and expert support and services to transform your applications with less effort.

You can also apply modern application development principles to your VMs and run all of your applications and workloads consistently across on-site datacenter, edge, and cloud environments. Increase developer productivity, simplify operations, and streamline infrastructure and application delivery to better support your business. With OpenShift Virtualization and Azure Red Hat OpenShift, you can meet today's business needs while preparing for future modernization and change.

Ready to migrate your virtual machines?

Unify and streamline IT operations with a single, cloud-ready application platform for VMs and containers.

OpenShift Virtualization and Azure Red Hat OpenShift reduce operational complexity and provide a unified, modern, cloud-native infrastructure for all of your virtualized and containerized applications and workloads. Plan your path to modernization with a platform and cloud services that bring modern application development principles to existing VMs and prepare your organization for a cloud-native future.

[Learn more](#) about OpenShift Virtualization.

[Read more](#) about Azure Red Hat OpenShift.

[Get started](#) with our self-paced online learning path for Azure Red Hat OpenShift.

Experience OpenShift Virtualization at no cost

The Red Hat Level Up Program gives eligible customers fully supported, no-cost access to Red Hat OpenShift, including OpenShift Virtualization, for their team, department, or organization initiatives for 1 year.

[Learn more](#)